Literature Survey: Reinforcement Learning for a Smart Thermostat Energy Saver

CSCE 642 – Reinforcement Learning, Course Project

Jessica Salguero (UIN 937005006) Omar Khan (UIN 437006822)

1 Summary of Current Project Plans

[finish after review]

1.1 Goal and research question

We are building a reinforcement learning (RL) controller for a simulated single-zone residential smart thermostat. The main question is: *can an RL agent learn anticipatory behavior, such as pre-heating or pre-cooling ahead of occupancy and peak electricity prices, that a reactive deadband controller cannot, and how does that behavior change as we shift the balance between energy cost and occupant comfort?*

1.2 Problem formulation

Environment. We implement a Gymnasium environment around a first-order resistor–capacitor (RC) thermal model,

$$T_{t+1}^{in} = T_t^{in} + \frac{\Delta t}{C}\left[\frac{T_t^{out} - T_t^{in}}{R} + Q_{\mathrm{HVAC}}(a_t)\right] + \varepsilon_t, \tag{1}$$

where R is thermal resistance, C thermal mass, $Q_{\mathrm{HVAC}}(a_t)$ the heating or cooling power of action a_t, and ε_t small random variation.

State. Indoor temperature (about 0.5°F bins for the tabular agent, raw float for DQN), outdoor temperature, hour of day (or a coarse time-of-day bucket), occupancy (home/away), day type (weekday/weekend), current HVAC mode, and a time-of-use price tier (off-peak/peak).

Actions. Off, low heat, high heat, low cool, high cool (five discrete actions). A continuous HVAC-power action is a stretch goal.

Reward. $r_t = -[\alpha E_{\mathrm{cost}} + \beta\,\mathrm{discomfort} + \gamma\,\mathrm{switch}]$, where discomfort is weighted more heavily when the home is occupied and may grow faster the further the temperature leaves the comfort band. The weights α, β, γ are the main experimental knob.

1.3 Methods, baselines, and evaluation

The primary agent is tabular ϵ-greedy Q-learning written from scratch. Baselines are a standard deadband (hysteresis) thermostat, which is the primary baseline, and a random-action agent as a sanity floor. Controllers are compared on total energy cost, average comfort violation, percentage of occupied time within the comfort range, and number of HVAC mode changes. Behavioral comparisons vary the outdoor-temperature pattern, weekday vs. weekend occupancy, peak vs. off-peak pricing, insulation quality (via R and C), and the reward weights. Stretch goals: DQN with continuous state inputs, historical weather data, continuous control (DDPG/PPO), multiple zones, uncertain occupancy, and a long-horizon energy comparison.

1.4 Changes since the proposal

[needs reveiw what to keep or not]

- **Scheduled-setback baseline.** The proposal's implementation plan (step 3) lists a scheduled-setback controller, but the baseline list in "Simplest First Results" does not. Related work (Sections 2.3) compares against programmable or static schedules, so we **propose making the scheduled-setback controller an explicit third baseline.**
- **Anticipation metrics.** To test the anticipatory-behavior hypothesis directly rather than infer it from aggregate cost, we **propose two additional metrics:** (i) the indoor-temperature deficit at the moment the resident arrives home, and (ii) the fraction of HVAC energy spent in the k timesteps before an occupancy or peak-price transition.
- **Comfort metric.** Following the observation in Sinergym [19] that percentage-of-time-out-of-range can misrepresent controller quality, we **report mean violation magnitude (degree-hours) alongside the percentage-of-time metric.**

2 Related Work

We organize the survey around five questions our project depends on: which RL algorithms we build on (Section 2.1), what is already known about RL for building control (Section 2.2), how comparable thermostat-level systems work (Section 2.3), how deep RL has been applied to HVAC (Section 2.4), and what simulators and thermal models are used (Section 2.5). Section 2.6 summarizes how our project differs.

2.1 Algorithmic foundations

Q-learning. Watkins and Dayan [1] prove that tabular Q-learning, the off-policy temporal-difference update $Q(s,a) \leftarrow Q(s,a) + \eta\,[\,r + \gamma \max_{a'} Q(s',a') - Q(s,a)\,]$, converges to the optimal action-value function provided every state–action pair is visited infinitely often and the step size η decays appropriately. Sutton and Barto [2] give the practical treatment we follow: ϵ-greedy exploration with a decaying ϵ, discussion of reward design, and model-based extensions such as Dyna that interleave real and simulated updates. Two consequences matter for us. First, the guarantee is asymptotic, so coverage of the table is the practical bottleneck. A naive product of our state features

(about 50 indoor bins × 10 outdoor bins × 24 hours × 2 occupancy × 2 day types × 3 modes × 2 price tiers) is roughly 2.9×10^5 states and 1.4×10^6 state–action pairs. This is an estimate from our proposed bin widths, and in practice many combinations are unreachable because occupancy and price tier are largely functions of (hour, day type). Coarse time-of-day buckets reduce the count by an order of magnitude, which is the trade-off the proposal's "debating 0–23 vs. buckets" note refers to. Second, including the current HVAC mode in the state makes the switching penalty Markovian, since the penalty depends on the previous action.

Deep Q-networks. Mnih et al. [3] replace the table with a neural network $Q_\theta(s, a)$ and stabilize training with experience replay (sampling decorrelated minibatches from a buffer) and a periodically frozen target network used to compute the bootstrap target. Their agent learns from raw pixels on Atari; ours would use a small multilayer perceptron on the continuous version of our state, with the same five discrete actions. This is exactly the proposal's DQN stretch goal, and the main comparison is sample efficiency and policy quality against the tabular agent.

Continuous control. DDPG [4] extends the DQN ideas to continuous actions with a deterministic actor $\mu_\phi(s)$ trained by the deterministic policy gradient through a critic $Q_\theta(s, a)$, using replay, slowly updated (soft) target networks, and additive exploration noise. PPO [5] is an on-policy method that constrains each policy update with a clipped surrogate objective, trading sample efficiency for stability and simpler tuning. Both are candidates for our continuous HVAC-power stretch goal.

Environment interface. Gymnasium [6] standardizes the `reset`/`step` interface, observation and action spaces, and the separation of `terminated` from `truncated` episodes. Our environment follows this API, which also lets us reuse off-the-shelf DQN/DDPG/PPO implementations for the stretch goals.

2.2 Surveys of RL for building control

Wang and Hong [7] review RL for building controls and break each study down by algorithm, state, action, reward, and environment. They find that only about 11% of the studies they surveyed involved real buildings, and name three barriers to adoption: training is slow and data-hungry, robustness and security of the controller need improvement, and generalization across buildings is weak (with transfer learning suggested as a remedy). Mason and Grijalva [8] survey RL for autonomous building energy management and trace HVAC applications back to 1996, covering Q-learning, hybrids of RL with model predictive control (MPC), neural fitted Q-iteration, and actor-critic methods such as A3C; they note, for example, the combination of Q-learning with Bayesian occupancy prediction by Barrett and Linder [10]. Nagy et al. [9] give a broader "ten questions" treatment aimed at buildings acting as flexible grid assets that must shift demand while maintaining comfort.

Relation to our work. These surveys supply the taxonomy we use to describe our design (state, action, reward, environment) and justify why we evaluate in simulation. We do not attempt deployment, so the real-building barriers they identify are out of scope, but the generalization barrier motivates our experiments across insulation levels and weather patterns. The surveys also show that discrete tabular agents are the early, simple end of the literature, which is the niche where a clean, fully inspectable study of anticipatory behavior is still feasible.

2.3 Thermostat-level and residential control

Barrett and Linder (2015) [10] are the closest prior work to our tabular agent. Motivated by commercial learning thermostats whose algorithms are undisclosed, they propose an architecture with two parts: a Bayesian learner that predicts room occupancy over time, and a Q-learning agent that learns the thermostat control policy, enabled by a purpose-designed state–action formalism. They report cost savings of about 10% over a programmable thermostat while keeping occupant comfort high. *Similarities:* Q-learning, a reward that trades off comfort against energy cost, and a thermostat-scale problem. *Differences:* occupancy is *predicted* in their pipeline, whereas we initially give the controller the true occupancy; their comparison is against a programmable thermostat, ours against a deadband controller (plus, if adopted, a setback schedule); and we add time-of-use pricing and a systematic sweep of reward weights. Their pipeline is effectively what our "uncertain occupancy" stretch goal would reproduce.

PreHeat (Scott et al., 2011) [11] is a non-RL system that senses occupancy, uses historical occupancy to estimate the probability of future occupancy, and uses that estimate to set the heating schedule automatically. It was deployed in five homes (three in the US, two in the UK) for about 61 days per house on average, alternating days between PreHeat and a static schedule and measuring gas use and "MissTime," the time the house was occupied but not warm. In UK homes PreHeat saved gas and reduced MissTime; in US homes it cut MissTime by a factor of 6–12 at similar gas consumption. *Relevance:* PreHeat is the strongest empirical evidence that anticipatory heating pays off in real homes, and MissTime is essentially the complement of our "percentage of occupied time in the comfort range." *Difference:* the anticipation is hand-engineered (prediction followed by a fixed rule), whereas we ask whether an RL agent *discovers* the same behavior purely from reward, and whether the learned lead time depends on the thermal time constant RC.

Ruelens et al. (2017) [12] apply batch RL, specifically fitted Q-iteration (FQI), to demand response with thermostatically controlled loads including a residential heat-pump thermostat. They extend FQI to settings where a forecast of exogenous data (such as price or outdoor temperature) is available, add a policy-adjustment step that injects expert knowledge, and also construct open-loop day-ahead schedules for market participation. Their experiments show batch RL is a viable alternative to model-based controllers without a system-identification step. *Similarities:* price-responsive control of a thermal load with a learned Q-function. *Differences:* FQI learns from a fixed batch of transitions and regresses Q with function approximation, while we learn online with a table; they feed forecasts into the state, and our hour-of-day and price-tier features play the role of a known exogenous forecast in a much simpler form.

2.4 Deep RL for HVAC control

Wei et al. (2017) [13] formulate building HVAC control as a Markov decision process and train a deep Q-learning controller in an EnergyPlus-based simulation, controlling the supply airflow rate for building zones; the data-driven approach avoids a detailed physical model at control time. This is the model for our DQN stretch goal. *Differences:* their setting is a multi-zone commercial-scale building simulated at high fidelity, whereas ours is a single-zone residential RC model with occupancy and price features.

Li et al. (2020) [14] cast data-center cooling as energy-cost minimization under temperature constraints and propose an end-to-end cooling control algorithm built on an off-policy, offline variant of DDPG. An evaluation network predicts an energy cost penalized by the cooling status of the

room, and a policy network outputs control settings given the current load and weather. On EnergyPlus they report about 11% cooling-cost saving relative to a manually configured baseline, and about 15% on a real trace from Singapore's National Supercomputing Centre under a 26.6°C inlet-temperature threshold. *Relevance:* a direct template for our continuous-control extension and for treating comfort as a constraint-like penalty. *Differences:* data-center load is driven by servers, not occupancy, and there is no price signal.

Gnu-RL (Chen et al., 2019) [15] targets deployability. The agent's policy is a differentiable MPC layer that encodes planning and linear system dynamics, is pre-trained by imitation learning on historical data from the existing controller (so it behaves reasonably from the start, hence "precocial"), and then improves online with a policy-gradient algorithm. It was tested in EnergyPlus and deployed on the airflow of a real conference room for three weeks. *Relevance:* Gnu-RL shows the benefit of baking physics into the policy for sample efficiency and interpretability. Our RC model in Eq. (1) is exactly a linear first-order dynamics model of the kind such a policy would fit, which makes a model-based variant (or Dyna-style planning in the tabular agent) a natural future comparison. *Difference:* our agent is model-free and learns from scratch in simulation, with no pre-training from an incumbent controller.

Xu et al. (2020) [16] study transfer learning for HVAC control, proposing a DQN-based method to reuse a trained controller across different building configurations and weather conditions to cut retraining cost. This speaks directly to the generalization barrier noted in [7]. *Relevance:* our planned experiments with better- and worse-insulated homes (different R and C) probe the same question at small scale: how brittle is a policy trained on one thermal configuration when the house changes? We do not implement transfer learning.

2.5 Simulation platforms and thermal models

Grey-box RC models. Bacher and Madsen [17] propose a procedure for choosing a model of a building's heat dynamics from a family of lumped-parameter (grey-box) models of increasing complexity, comparing them with likelihood-ratio tests. These models are written as stochastic differential equations whose parameters (thermal resistances and capacities) have direct physical meaning. Our single-node model, Eq. (1), is the simplest member of this family: it has time constant $\tau = RC$, relaxes toward $T^{out} + R\,Q_{\mathrm{HVAC}}$ under constant power, and its additive noise ε_t plays the role of the diffusion term in the stochastic formulation. Because we discretize with forward Euler, we need $\Delta t/(RC) \ll 1$ for the discrete model to track the continuous one, a constraint to check when choosing Δt. The steady state $T^{out} + RQ$ also tells us whether "low heating" can physically reach the comfort band in cold weather, which constrains how we should set Q for each action. Unlike [17], we do not identify parameters from data; we set them by hand to represent houses of different quality, which is a limitation we will state in the report.

EnergyPlus-based testbeds. Moriyama et al. [18] release an open-source wrapper that connects EnergyPlus to RL frameworks and use it to train deep RL on a two-zone data-center model, reporting a 22% improvement over the built-in model-based controller. Sinergym [19] is a more recent Gymnasium-compatible testbed that connects to EnergyPlus through its Python API, supports discrete and continuous actions, interchangeable weather files (with Ornstein–Uhlenbeck noise to vary weather between episodes), and custom reward functions. Its default reward is a weighted combination of energy and comfort violation, and a "schedule" reward varies the discomfort weight with occupancy, which closely matches our occupancy-weighted discomfort term. In its example study, deep RL controllers (PPO, TD3, SAC) saved roughly 15–18% energy over reactive controllers

in several months, and PPO was the most stable. The authors also observe that percentage of time outside the comfort range can be a misleading metric compared with the average violation magnitude. *Relation to our work:* these platforms offer high fidelity at a high simulation cost, and their authors note that building simulators are expensive to run and awkward to wrap in an RL loop. Our lightweight RC environment gives up fidelity for millions of cheap timesteps, full control over occupancy and price, and complete reproducibility, which suits a controlled study of how behavior changes with reward weights. Sinergym's reactive rule-based baselines are the analogue of our deadband controller.

Comfort range. We define the resident's comfort band from standard thermal-comfort guidance [20]; the band itself is a tunable parameter of the reward and evaluation rather than a modeling claim.

2.6 Synthesis: how our project differs

Table 1 summarizes the comparison. Taken together, the prior work establishes three things: RL can beat reactive and scheduled controllers on cost at comparable comfort [10, 14, 18, 19]; occupancy prediction and pre-conditioning are valuable in real homes [11]; and the main open problems are sample efficiency, robustness, and generalization [7, 15, 16]. Our contribution is deliberately narrow and educational rather than state of the art:

1. a transparent, reproducible, low-cost residential environment that includes occupancy, time-of-use pricing, and a switching penalty, so that anticipation is *possible* and *measurable*;
2. a study of whether anticipatory behavior emerges from a tabular agent and how it depends on the cost/comfort weights (α, β, γ) and on the thermal time constant RC, rather than only reporting aggregate savings;
3. an inspectable policy (a Q-table that can be queried by hour, occupancy, and price) against which a DQN can optionally be compared.

We do not claim to improve on the methods above, and we make no claim about real-building deployment.

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Work	Method	Setting	Key difference from ours
Barrett & Linder [10]	Q-learning + Bayesian occupancy prediction	Thermostat, simulated	Occupancy predicted, not given; no price tiers; programmable-thermostat baseline
PreHeat [11]	Occupancy prediction + schedule (no RL)	Five real homes	Hand-built anticipation; we test whether RL learns it
Ruelens et al. [12]	Fitted Q-iteration (batch)	Heat pump, water heater	Batch regression with forecasts vs. our online table
Wei et al. [13]	DQN	EnergyPlus building	Multi-zone, high fidelity; no occupancy/price focus
Li et al. [14]	Offline DDPG	Data-center cooling	Server load, no occupancy or price
Gnu-RL [15]	Differentiable MPC + policy gradient	EnergyPlus; real room	Model-based, imitation pre-training; ours model-free from scratch
Xu et al. [16]	DQN transfer learning	Varied buildings/weather	We only test robustness to R, C changes
Sinergym [19]	PPO/TD3/SAC on EnergyPlus	Data-center example	High-fidelity platform; ours is a lightweight RC model

Table 1: Comparison of the most closely related systems with the proposed project.

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